

THE ART AND SCIENCE OF LUCAS FILM: WE BUILD COOL STUFF A DAY IN THE LIFE OF A LUCASFILM SOFTWARE DESIGNER

CLASSROOM GUIDE

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SUMMARY

Mike Jutan, Rachel Rose, and Matthew Parrott talk about their work as software engineers in Lucasfilm's Research & Development department. They will discuss how their experiences studying both computer science and art helped them to achieve their goals of working as software engineers for Lucasfilm.

Grades 6-12

Suggested Subject Areas

Arts/Media
Career Advisory

Computer Science
Language Arts

Physics

All SFFS Youth Education materials are developed in alignment with California educational standards for media literacy.

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SAN FRANCISCO
FILM SOCIETY

PRESENTERS

Matthew Parrott works as a software engineer at Lucasfilm. As a practicing fine artist, his interests weave together a love of art and science with the belief that people who work together can help one another grow. Matthew began training for a career in visual effects during high school and has worked on films such as *Cars*, *Avatar*, and *Pirates of the Caribbean IV*. His art has been shown internationally and in collaboration with artists such as Patrick Dougherty, Peter Foucault, and Don Foresta. Matthew's various software projects include work for Mars rovers, video games for kids who are afraid of snakes, molecular field doohickies, and virtual menorahs. Additionally, he volunteers at 826 Valencia, teaching workshops on filmmaking and computer art.

Matthew's gallery:
<http://www.matthewatabet.com>

Peter Foucault:
<http://peterfoucault.wordpress.com/>

Malcolm Kesson's Renderman Tutorials:
<http://www.fundza.com/>

Rhino 3D, Free download:
<http://www.rhino3d.com/download.htm>

Houdini, Free download:
<http://www.sidefx.com/download>

Rachel Rose is a Research & Development Engineer at Lucasfilm who specializes in developing tools that make animating onscreen characters easier. Her passion for learning led her to earn a B.A. in Computer Science and English from Grinnell College and an M.S. and Ph.D. in Computer Science from the University of Wisconsin-Madison. Rachel's interest in movies, television, and video games makes Lucasfilm a perfect fit for her professional and personal motivations.

Personal Website:
<http://pages.cs.wisc.edu/~heckr/>

Computer Animation:
<http://entertainment.howstuffworks.com/computer-animation.htm>

Mike Jutan is an R&D Engineer at Industrial Light & Magic in San Francisco, California. A graduate of the University of Waterloo's Computer Science Co-op Program, Mike worked as a Software Developer for Maya at Alias in Toronto, Canada. He gained production experience at Pixar Animation Studios as both a Production Engineer and a Modeling/Shading TD before moving to ILM in 2007. Mike recently spoke at the TEDxIB@York conference in Toronto, and has taught technical classes for Lucasfilm's TV, Film and Video Game divisions in San Francisco, Marin County, and Singapore.

More about Mike and his work:
<http://www.mikejutan.com>

Check out Mike's TEDx talk here:
<http://www.youtube.com/watch?v=4I5J4P0XaPA>

ACTIVITIES & DISCUSSION

POST-PRESENTATION DISCUSSION

Did you enjoy this presentation? What were your favorite moments?

Did you enjoy the Lucasfilm software developers who presented? What more would you like to know about them? What are some of your favorite movies that they've worked on?

Enthusiasm and passion are important parts of these professionals' lives. How do you feel about the message of enthusiasm in the presentation and links? How vital do you think it is to success?

How has enthusiasm impacted the lives of the presenters?

Have you ever found enthusiasm to be critical to your own success in a situation?

What are the obstacles to enthusiasm? Is enthusiasm always sustainable?

What types of activities, hobbies, careers, sports, etc. are you enthusiastic about?

If you had to enter the workforce today and could choose any field, but would be paid the same salary regardless of your choice, what field would you choose?

Each of the presenters achieved lifelong goals. Aside from enthusiasm, what qualities do you think helped each of them succeed? How did they start out? What skills did they develop?

What did you know about computer animation

or software development before attending this presentation? What did you learn? What impression did you expect to have of the presenters based on their profession? How did it differ from your actual impression?

Do you think that software development, filmmaking or computer animation would be a rewarding career? Why or why not? Did this presentation change your interest level at all?

LANGUAGE ARTS

"Strive to be happy" is one of Mr. Jutan's favorite lines, from the 1927 Max Ehrman poem *Desiderata* (see below). What does the word "strive" mean to you in this context? Is it a better word choice than "intend" or "try" would have been? Why or why not?

Desiderata

Go placidly amid the noise and haste,

and remember what peace there may be in silence.

As far as possible, without surrender, be on good terms with all persons.

Speak your truth quietly and clearly; and listen to others,

even to the dull and ignorant; they too have their story.

Avoid loud and aggressive persons, they are vexations to the spirit.

If you compare yourself with others, you may become vain and bitter,

for always there will be greater and lesser persons than yourself.

Enjoy your achievements as well as your plans. ►

► Keep interested in your own career, however humble;
it is a real possession in the changing fortunes of time.
Exercise caution in your business affairs,
for the world is full of trickery.
But let this not blind you to what virtue there is;
many persons strive for high ideals,
and everywhere life is full of heroism.

Be yourself. Especially do not feign affection.

Neither be cynical about love;
for in the face of all aridity and disenchantment
it is as perennial as the grass.

Take kindly the counsel of the years,
gracefully surrendering the things of youth.

Nurture strength of spirit to shield you in sudden
misfortune.

But do not distress yourself with dark imaginings.
Many fears are born of fatigue and loneliness.

Beyond a wholesome discipline, be gentle with
yourself.

You are a child of the universe no less than the trees
and the stars;

you have a right to be here. And whether or not it is
clear to you,

no doubt the universe is unfolding as it should.

Therefore be at peace with God, whatever you conceive

Him to be.

And whatever your labors and aspirations,
in the noisy confusion of life, keep peace with your
soul.

With all its sham, drudgery and broken dreams,
it is still a beautiful world.

Be cheerful. Strive to be happy.

Max Ehrman

SCIENCE

Why is knowledge of physics important to a computer animator? What about anatomy/biology? What about geology or astronomy? Is a background in science essential to a career in animation or software development?

ARTS/MEDIA

Additional Computer Animation Resources

Autodesk offers free versions of CGI software Maya and MotionBuilder to education community members.

You may also view a demo version and see classroom examples of SAM Animation, a stop-frame system developed at Tufts University's Center for Engineering Education and Outreach, at iCreate to Educate. Softronic offers several free programs, including the popular Pivot stick figures animator.

POST-PRESENTATION ACTIVITY

RESEARCH YOUR DREAM JOB

An assignment relished by Mr. Jutan in his youth; each student spends 30 minutes researching the practical aspects of achieving his or her "dream job" or at least one that holds the most interest. Students should try to ascertain:

1) The types of academic knowledge useful to that profession

2) The educational qualifications typical to that profession

3) Any ancillary qualifications typical to that profession

4) The specific regions where that profession is generally conducted (geographic, demographic, urban/rural, etc.), if applicable

5) The means by which companies in that field conduct their hiring

6) The specific types of jobs available in that field, if applicable

7) Bios/histories of the most successful individuals in that profession



THE ART & SCIENCE OF LUCASFILM

This program is a dynamic collaboration between the Film Society and legendary film and entertainment company Lucasfilm. In this yearlong series, experienced professionals from the various Lucasfilm divisions share their knowledge—and their spectacular secrets—with Bay Area students in engaging and interactive multimedia presentations that demonstrate the intersection of art and science in the entertainment industry.

The program is designed to complement a range of high school curricula, including arts, language, social studies, math and sciences. Each event allows time for discussion, where students have the opportunity to ask questions of Lucas directors, supervisors, artists and engineers. The series aims to inspire future generations of filmmakers, storytellers, engineers, and artists; demonstrate the range of career opportunities available in today's digital entertainment industry; and give an insider's view of today's popular TV, film and video game productions.

Past Presentations

Inside the Adventure: How LucasArts Games Are Made

Art director Matt Omernick and software engineer Eric Johnston provide a close-up look at what LucasArts puts into each game project. With an inside view of the art as well as the technology, they demonstrate how early drafts and concepts become a final, polished experience. Matt and EJ give expert under-the-hood detail of the prototyping process they use and show how students can create games of their own.

Lucasfilm Animation

Dave Filoni, supervising director of the animated film and TV series *Clone Wars*, gives a behind-the-scenes look at the newest division of Lucasfilm, Lucasfilm Animation.

Skywalker: Creating Sound for Film

Will Files, sound designer of films such as *Cloverfield*, *Bee Movie*, and *Madagascar 2*, provides an in-depth look at the art and craft of sound in cinema. The presentation explores the process of how a feature film soundtrack is created, the impact that sound can have on cinematic storytelling, and the choices that are made along the way. It includes interactive demonstrations of sound creation, as well as clips from both recent projects and classics from Skywalker Sound's long history as the leader in motion picture sound.

ILM: Inside the Art Department

Industrial Light & Magic art directors David Nakabayashi, Aaron McBride, and Christian Alzmann take students on a journey through the past, present and future of the pioneering visual effects company. The presenters discuss their early education and interest in art and film, how they got their start at ILM, and how their artistic training plays a major role in the creative process, from concept art to design and art direction.

LUCASFILM OVERVIEW

Founded by George Lucas in 1971, Lucasfilm is a privately held, fully integrated entertainment company. While best known for its association with the *Star Wars* and *Indiana Jones* film franchises, Lucasfilm is actively involved in many facets of the entertainment industry, operating seven different divisions that lead in their respective fields.

Lucasfilm is responsible for the production, promotion and strategic management of the company's various theatrical, television and entertainment companies. These include *Star Wars*, the *Indiana Jones* trilogy, *The Adventures of Young Indiana Jones*, *THX 1138*, *Willow*, *Labyrinth* and more.

Industrial Light & Magic, better known as ILM, has set the standard for visual effects, creating some of the most stunning images in the history of film since its founding in 1975 by George Lucas. ILM houses the leading effects facility in the world, one that has served the motion picture industry for 35 years. It's been awarded 15 Academy Awards for Visual Effects, 23 Scientific and Technical Achievement Awards, and played a key role in 10 of the top 15 worldwide box office hits of all time.

LucasArts is a leading developer and publisher of interaction entertainment software for videogame console systems and personal computers. LucasArts was created by George Lucas in 1982 to provide an interactive element to his vision of a state-of-the-art, multifaceted entertainment company. Its games continually receive critical acclaim and attain commercial success, with numerous Game of the Year honors and multiple awards from the Academy of Interactive Arts & Sciences.

Lucasfilm Animation is a digital animation studio producing content for global audiences, including television, video games and visual effects. It was founded in 2003 while its sister office, Lucasfilm Animation Singapore, was founded in 2004. Current projects include *Star Wars: The Clone Wars*, a computer-generated television series combining legendary Lucasfilm storytelling with cutting-edge animation and signature style.

Skywalker Sound serves the digital audio post-production needs of the entertainment industry. A dynamic force in the art of sound design, editing, and mixing for more than 25 years, its work has been honored with 18 Academy Awards. Skywalker Sound has provided sound services for more than half of the top 25 highest-grossing films of all time.

Lucas Licensing is responsible for licensing and merchandising activities relating to *Star Wars*, *Indiana Jones*, and other Lucasfilm entertainment properties. With over \$8 billion in worldwide consumer sales, Lucas Licensing manages a program that includes the best-selling boys' action toys of all time; merchandise sold in over 100 countries; and the prestigious *Star Wars* museum exhibitions, attended by more than 3.5 million visitors.

Lucas Online was founded in 1997 as a digital destination for Lucasfilm properties. It has pioneered entertainment on the web in areas such as video streaming, web documentaries, interactive content, insider access to Lucasfilm's creative teams and daily news updates. Current Lucas Online projects: starwars.com, starwarshop.com, lucasartstore.com and thx1138movie.com.